POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Basic Electrical Engineering (New) | Pass Marks : 45
Time : 3 hrs.

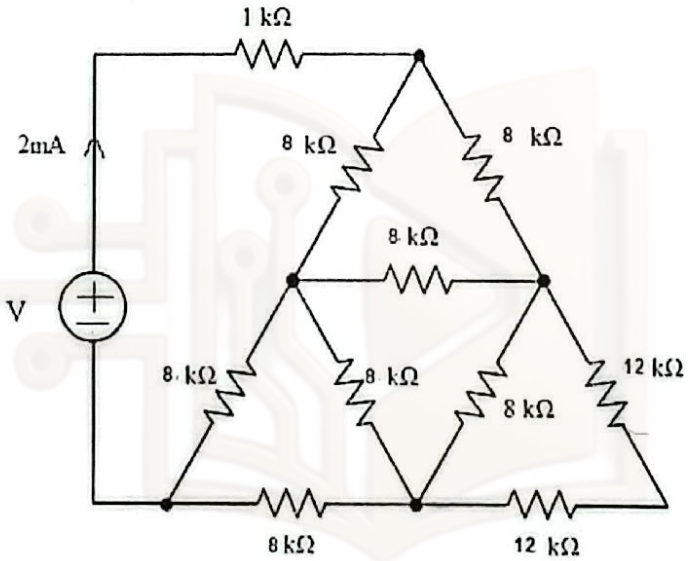
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

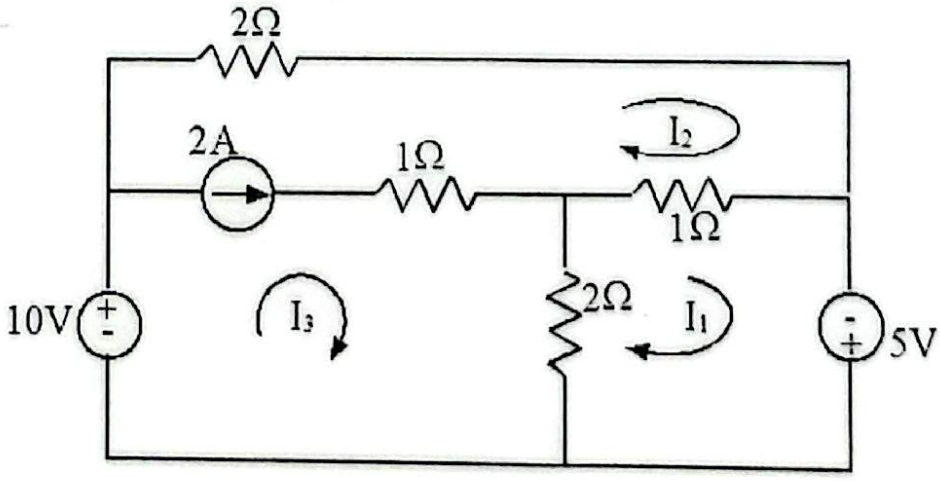
Attempt all the questions.

1. a) With the help of single line diagram, explain generation, transmission and distribution of electricity in Nepal. 7

   b) Find equivalent resistance and Voltage V. 8

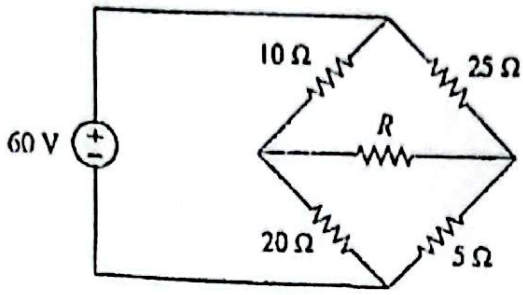


2. a) Find the mesh current I_1, I_2 and I_3 in the circuit given below using the concept of mesh analysis. 7



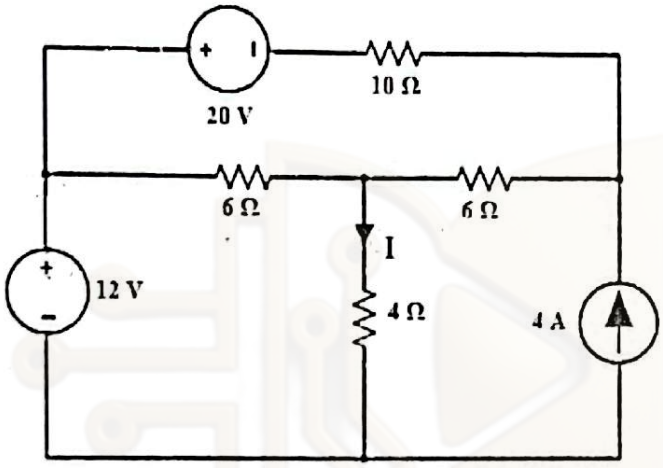
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b) Calculate the value of R that will absorb maximum power from the circuit. Also calculate maximum power absorbed by R. 8

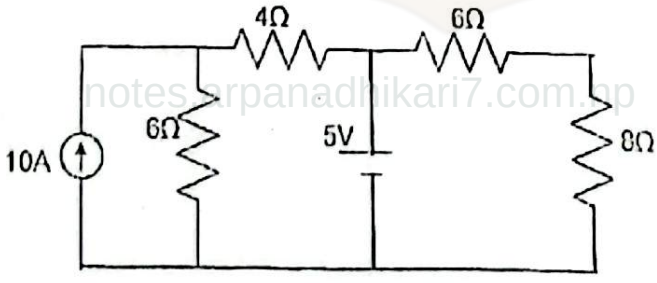


OR

State Superposition Theorem and Find current through 4 Ω using Superposition Theorem.

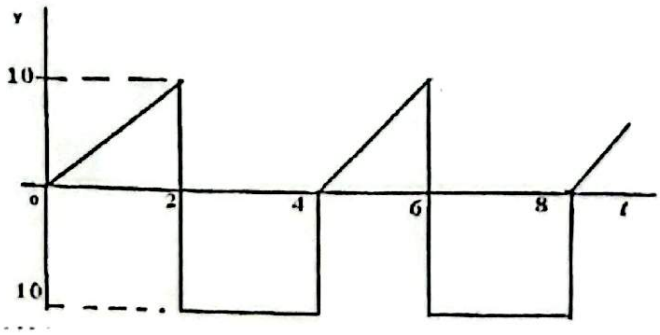


3. a) Calculate the power dissipation across 4 ohm resistor using Thevenin's theorem. 7



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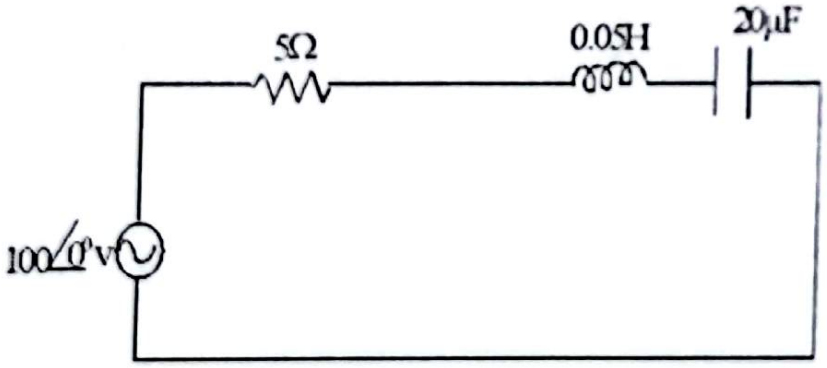
b) Calculate the average and RMS value of voltage for waveform as shown in figure below. Also calculate form factor. 8



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4. a) Two impedances (150+j15) Ω and (110-j80) Ω are connected in parallel across a 330V, 50Hz supply. Find active power, reactive power, apparent power and power factor of the circuit. 7

b) In the series circuit shown, the applied voltage is 100 V of variable frequency. If the frequency is varied till the resonance occurs in the circuit than find the current, resonant frequency, bandwidth, quality factor and the voltage across the inductors and capacitors. 8



5. a) Three equal impedances having resistance 32 Ω and inductive resistance 10.5 Ω are connected in delta connected system. Find: 8

i. Phase current
ii. Line current
iii. Power factor
iv. Power consumed

b) Explain the operating principle of DC shunt generator. Also, find the expression for the generated emf by DC shunt generator. 7

6. a) Explain the construction and operating principle of three phase induction motor. 7

OR

Explain the construction and operating principle of DC motor.

b) The following results were obtained on a 4 kVA, 200/400 V, 50 Hz transformer. 8

Open circuit test: 100W, 1A, 200V

Short circuit test: 85W, 10A, 15V

Determine equivalent circuit parameters referred to LV side.

7. **Write short notes on: (Any two)** 2×5

a) Electric circuit vs magnetic circuit
b) Short circuit and open circuit
c) Single phase vs three phase system

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